

B1-RH-ENUM claim-level synthesis: Reading-H vacuum selection, from the enumerated base to the C_full theorem

TECT verification-first repository · claim B1-RH-ENUM

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1. Purpose and scope

Claim-level synthesis for B1-RH-ENUM: the parent document of the claim’s sub-proof units (taxonomy of governance/claims-restructure-proposal-260609.md). It adds no mathematics; every statement carries the unit’s registered tier. **Scope:** the composition of B1’s units into the canonical claim state. **Not claimed:** anything beyond the head theorem’s enacted scope (T7-SCOPE_{C_{full}} given A1, window $\mathcal{R}$); no A1-free, global, or all-intensity statement.

2. The claim and its canonical state

Claim (canonical). Within $\mathcal{R} = \{\mu^2 \in (0, 0.0342), I \in (0, I_c^{\text{sel}}(\mu^2))\}$, given the A1 kernel convention, $F[Q] - F[G_*] > 0$ for every admissible competitor $Q \in \mathcal{C}_{\text{full}}$ (real antipodal, $|Q| \leq n_{\text{pack}}$, pairwise geodesic $\geq \theta_{\text{min}}$). Tier: T7-SCOPE_{C_{full}} (enacted 2026-06-10, comfortable off-diagonal $\times 1.96$); hypothesis: A1-KERNEL-CONV; open gates: none.

3. Sub-proof unit map (per-unit tier and role)

Unit	Tier	What it contributes	Publication state
Reading-H/	T7-SCOPE	the HEAD: (D)(O)(S) assembly + window certification (res5_032-036)	PUBLISHED – Reading-H-cFull-T7-260611 (27/27)
enumerated/	T5	the numerical base: enumerated single/two-shell ensembles, 167/167 asserts, migration re-validated	AUXILIARY (provenance record; operator-accepted draft)
ESTIMATOR-UPGRADE/	T4	controlled-error margins for the enumerated estimator (GAP-2 instance)	AUXILIARY (cited, not load-bearing)
ROBUSTNESS-MU2/	closed band	off-anchor robustness: $\mu^2 \in [\times 0.5, \times 2]$, worst margin $\times 2.55$	AUXILIARY (operator-accepted draft)
near-gap/	aux	convention-exact closure + the self-caught $M' = -J(0)/2$ factor-2 retraction	AUXILIARY (operator-accepted draft)

External load-bearing inputs (other claims): Prop-A (B2, T6 on $\mathcal{A}_{\text{adm}}$) for (D)/(O) on the primary class; DR-2 lattice (B5, T7 NT-standard) backing Lemma 1; STEP-5B rectangle (B5, T6) for the K-budget; SC-SCOPE (B5, T5 thin) for (S). Full per-edge accounting: parent capstone Main-Line-Synthesis-T013-260612 Sec. 3–4.

4. Composition and what remains

The head package proves (D)(O)(S) competitor-agnostically inside $\mathcal{C}_{\text{full}}$; the enumerated base (T5) anchors the numerics; the auxiliary units supply robustness, controlled-error and convention-exactness corroboration without entering the analytic chain. The claim’s canonical tier is therefore carried ENTIRELY by the head package plus its named external inputs – consistent with the claim card’s empty gate list.

Remaining tracks touching B1 (none blocking): T-030 (arbitrary- Q DR-2, frontier; NOT load-bearing – Lemma 2 caps $T' \leq 10$ in-class), T-031 (STEP-5B exhaustiveness decision layer, B5-side), T-006 (code de-hardcoding, infra).

5. Devil’s-advocate / self-adversarial review (CLAUDE.md 6.3)

- (α) “The claim NAME says ‘enumerated ensembles’ but the canonical tier is a C_{full} statement.” VALID with mitigation: the claim grew from the enumerated T5 base (2026-06-05) to the C_{full} head (2026-06-10); the name is historical. The canonical statement is the status.json scope; flagged for an operator rename decision (cosmetic).
- (β) “The claim.md header still reads ‘Tier: T5, last review 2026-06-05’.” VALID – a SYNC DEFECT, not a status question: status.json (T7-SCOPE) is the single source of truth and CLAIMS.md agrees; the stale claim.md header is registered here for the next claim-card edit. No status is derived from claim.md.
- (γ) “Synthesis could smuggle the auxiliary units into the load-bearing chain.” DISMISSED: Sec. 3 marks every auxiliary unit AUXILIARY and the analytic chain runs head + external inputs only, exactly as in the parent capstone.

Quantitative sanity check (mandatory). The unit tiers quoted here match the generated INDEX.md and the five bundle footers; the head’s numbers ($R_{\text{lead}} = 0.510$, joint ≥ 1.040 , window $\mu^2 < 0.0342$) are those of the published bundle, reproduced by res5_032–036 (27/27).

6. Result footer

Result ID:	B1-RH-ENUM-Synthesis-T013-260612 (PUBLISHED claim-level SYNTHESIS, T-013)
Precise statement:	B1’s canonical state (T7-SCOPE_{C_full} given A1, window R, gates none) is carried by the PUBLISHED head package Reading-H-cFull-T7-260611 plus named external inputs (Prop-A T6, DR-2 T7-lattice, STEP-5B T6, SC-SCOPE T5); the in-claim units enumerated/ESTIMATOR-UPGRADE/ROBUSTNESS-MU2/near-gap are auxiliary corroboration, not load-bearing.
Scope:	synthesis layer; no new math, no tier change, no gate flip.
Evidence grade:	SYNTHESIS over PUBLISHED + operator-accepted units.
Reproduction command:	run the Reading-H bundle entry manifest (res5_032--036); auxiliary units per their own notes.
Falsifier:	any falsifier of the cited footers; or a unit-tier mismatch against INDEX.md.
Tier before / after:	no change (B1 stays T7-SCOPE_{C_full}).
No-overclaim statement:	nothing beyond the head’s enacted scope; the claim-name/scope drift and the stale claim.md header are flagged, not papered over.
Next required action:	operator: optional claim rename decision + claim.md header sync at next card edit.